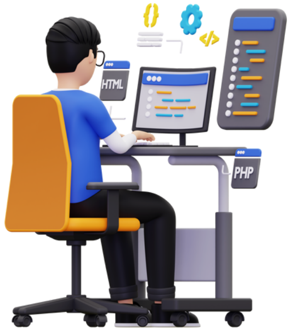
Appleton Innovations

Internet of Things Developer

Embark on a journey to become a **Certified Internet of Things (IoT) Engineer** with this program. You'll gain a strong grasp of creating and putting into action IoT solutions using devices connected to sensors and an IoT gateway. Plus, you'll dive into the nitty-gritty of setting up an IoT Cloud to store and analyze data. This course is designed by industry experts and is packed with hands-on activities and real-world examples to make learning engaging. You'll get to explore the world of IoT using tools like Arduino, Raspberry Pi, Python, Linux, AWS, Azure, and ThingsBoard Cloud. Get ready to dive into the exciting realm of IoT!

Python Programming



Comprehensive Python Training Course will teach students the basics of Python, data operations, conditional statements, shell scripting, Flask, Data Analysis. This Python course will give students hands-on development experience and prepare them for an exciting career as a python developer.

Introduction to Python

- Need for Python Programming
- Advantages of Programming
- Overview of Python
- Organizations using Python
- Python Applications
- Python, Jupyter installation
- Learn to use Jupyter Note Book
- Variables, Operands and Expressions

Python Data Types

- Declaring and using Numeric data types
- Using string data type and string operations
- Use of List, Tuple data type
- Tuple - Properties, Related Operations
- List - Properties, Related Operations

Python Program Flow Control

- Conditional blocks using if, else and elseif
- Simple For loops in python
- For loop using ranges, string, list and dictionaries
- Use of while loops in python

Python Exception Handling

- Avoiding code break and
- Safeguarding file operation using exception handling
- Handling and helping developer with error code
- Programming using Exception handling

Python String, List And Dictionary Manipulations

- Building blocks of python programs
- Dictionary manipulation
- Programming using string, list and dictionary in build functions
- Dictionary - Properties, Related Operations
- Set - Properties, Related Operations

Python File Operation

- Reading config files in Python
- Writing log files in Python
- Understanding read functions, read(), readline() and readlines()
- Understanding write functions, write() and writelines()
- Manipulating file pointer using seek Programming using file operations

Python Functions, Modules And Packages

- Organizing python codes using functions
- User-Defined Functions
- Function Parameters
- Different Types of Arguments
- Global Variables and Keywords
- Variable Scope and Returning Values
- Lambda Functions,Built-In Functions
- Modules and Packages

Deep Dive – Functions and OOPs Topics

- Introduction to OOPs
- Built-In Class Attributes
- Public, Protected and Private Attributes, and Methods
- Class Variable and Instance Variable
- Constructor and Destructor
- Decorator in Python
- Core Object-Oriented Principles

course projects

- Contact Book Project
- Dice Rolling Simulator using Python

Internet of Things using Raspberry Pi

The course focuses on Linux based operating systems, advanced networking, user interfaces and uses more computing intensive IOT applications as examples. You will be working on Raspberry Pi, Sensors and will gain hands-on experience with AWS IoT Core, Blynk Local Server and other cloud platforms. This course will have real time projects and case studies which will help you master the Internet of Things concepts.

Getting Started with Raspberry Pi

- Introduction to Raspberry Pi?
- Powering Raspberry Pi
- Installing OS on your Raspberry Pi
- Computer using Raspberry pi
- Raspberry PI Configuration
- IP Address of Raspberry
- Starting up with Software
- Software available in Raspbian OS
- Linux Commands
- File System in Raspberry PI
- Working with text
- Nano editor, Writing and executing programs
- Introduction to Virtual Network Computing
- VNC Installation and Raspberry Pi GUI access
- Remote file transfer

Interfacing Hardware with the Raspberry Pi

- Simple LED control using bash commands
- LED Blinking using Python Raspberry pi Library
- Temperature and Humidity sensing using DHT-11 sensor and Raspberry Pi
- Motion detection using Raspberry pi
- Button Interfacing with Raspberry Pi
- Object Detection using Raspberry Pi
- Interfacing LCD display
- Interfacing Relay module
- MCP3008 ADC Converter
- Interfacing LDR, thermistor Analog temperature sensors

IoT with AWS IoT Core

- Introduction to AWS IoT Core
- Connecting things to AWS IoT Core
- Sending Notifications
- Data Visualization using AWS Quicksight

Cloud Data Monitoring using Raspberry Pi

- IoT using Thingspeak Platform overview
- Create Account in Thingspeak
- Thingspeak (HTTP) REST API
- Sending sensor data to Thingspeak Cloud
- Sending Notifications or Alerts

Node-Red: M2M and Gateway

- Introduction to Node-RED
- Getting started with Node-Red
- Node-Red Basic Nodes
- Node-Red Dashboard
- HTTP & MQTT Nodes

GPIO Control using Python and Flask Framework

- Build a basic Flask website
- Add a new page
- Html page
- Add a HTML template page
- Add colour with CSS
- Flask and Python based Web Application to control GPIO over Internet
- Flask and Python based Web Application to display sensor data

course projects

- Smart City Traffic Management Solutions
- Smart Industrial solution

IoT using Arduino

This course deals with basic electronics, microcontroller architectures, sensors, basic networking, IoT protocols and IoT Cloud Platforms. Using IoT kit and Arduino Platform, you will develop projects like Home Automation using IOT, Weather Station; IOT based smart irrigation controller and many other projects.

Introduction to Internet of Things

- What is IoT?
- How IoT Works?
- Purpose of IoT
- IoT Architecture
- History of IoT
- IoT Potential

Getting Started with Arduino

- Installing Arduino IDE
- Arduino Programming, Arduino IDE
- ESP32 IoT Board
- Installing ESP32 Board in Arduino IDE
- Programming ESP32 Board
- Blinking LED Project

Sensors, Actuators & Electronics

- Working with Buzzer
- Basic Electronics, Circuit Design
- Traffic Light Control System
- Producing Colors by using RGB Led
- Analog Sensors, LDR sensor
- LM35 Temperature sensor
- Measuring Temperature and Humidity using DHT11 sensor
- Working with Push Button, IR sensor

Wireless Communication Technologies

- Wireless Communication Protocols
- Bluetooth, WiFi Technology
- RFID,, Zigbee Technology
- Cellular, LoRa, NB-IoT, Sigfox Communication
- TCP-IP Protocol
- ESP32 WiFi Communication
- WiFi Scanning Networks
- WiFi Network Communication

IoT using Blynk Mobile Platform

- Blynk IoT Platform
- Blynk APP
- Blynk Project Creation
- Home Automation Project
- Smart Bulb project
- Blynk Virtual Pins
- Sensor data Visualization
- Blynk Notifications

Smart Applets using IFTTT

- Introduction to Smart Applets
- IFTTT
- Smart Applet to Send Weather data to Email
- Voice Assistant with Google docs
- Weather Data Logging in Google sheets
- Voice Controlled Home automation
- Notification Cube
- Automation using Zapier

Cloud Data Monitoring using Arduino

- Thingspeak IoT Cloud Overview
- Thingspeak Account Creation
- Thingspeak API
- Sending Light Values to Thingspeak Cloud
- Sending Temperature and Humidity Values to Thingspeak Cloud
- Visualizing Sensor Data in APP
- Sending Alerts or Notifications

ML in IoT using MATLAB

- Introduction to Matlab
- Data Analysis with Matlab
- Data Visualization using Matlab
- Machine Learning with IoT Data

course projects

- Smart Street Lighting System
- Smart Home Solutions
- Smart Shoe Project

IoT Cloud Platforms

IoT is considered to be an essential part of the Industry 4.0 along with AI and Big Data. This course deals with the rapidly evolving field of cloud computing. We will examine cloud offerings from all the leading providers, including Microsoft Azure, Amazon Web Services and Google and how they can be used in developing IoT applications.

Cloud Platforms for IoT

- Introduction to IoT and Cloud
- Cloud as PaaS, SaaS, IaaS
- Use of Cloud in IoT

Microsoft Azure Cloud Platform

- Introduction to Microsoft Azure IoT Architecture
- Creating Microsoft Azure free account
- Connecting ESP32 to Azure IoT Hub
- Sending User Notifications
- Visualizing the Sensor data

Amazon Web Services IoT

- Introduction to AWS IoT Architecture
- Creating Things and Policy
- Connecting ESP32 to AWS IoT Core
- Storing IoT Data
- Data Visualization
- Sending Notifications

Google Cloud IoT

- Google Cloud IoT Core Architecture
- Getting started with IoT Core
- Connecting ESP32 to Google Cloud
- Sending Data to Google Cloud
- Sending Notifications
- Visualizing sensor data

course projects

- Smart Cold chain solution using any cloud platform

IoT Data Analytics (Optional)

In this course, you will learn how to collect and store data from a data stream. You will prepare IoT data for analysis, analyse and visualize IoT data. You will implement a machine learning model to take action when certain events occur and deploy this machine learning model.

Introduction to IoT Analytics

- Data Science
- IoT Analytics and Case studies

Python for Data Analysis and Machine Learning

- Data Analysis using NumPy, SciPy, Pandas
- Data Visualization using Matplotlib and Seaborn
- Machine Learning using Sci-kit Learn

Accessing IoT Data

- Data Acquisition
- HTTP protocol to collect data
- MQTT protocol to collect data
- Datasets creation

Processing and Analyzing IoT data

- Perform EDA, Data Visualization
- Missing Data, Time Series data
- Correlation: Heatmaps, Pairplot, Outliers
- Seasonality and Trends

Deploying ML Models

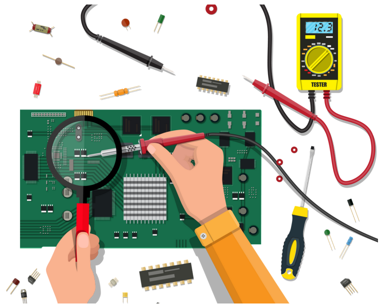
- Machine Learning Algorithms
- Prepare data for machine learning
- Split Training and testing data
- Machine Learning Algorithms
- Prepare data for machine learning
- Split Training and testing data

course projects

- Smart Irrigation based on sensor data
- Energy optimization using outlier detection

PCB Designing and Soldering Course (Optional)

This is a basic course for designing PCB using the software. PCB (Printed Circuit Board) designing is an integral part of each electronics product and this program is designed to make students capable to design their own projects PCB up to industrial grade.



PCB Design Basics

- PCB Design basics
- Introduction to EDA Softwares
- PCB Design using EASYEDA
- Creating New Project, Schematic Editor
- Adding Components to Schematic
- Schematic to Board Layout
- Snap Grid & Ratlines, Component Placement, Manual routing
- 3D Viewer, Placing a via, Auto routing
- Mounting holes and cutouts
- Copper Pour, Adding Text and Images
- Generating GERBER & BOM

PCB Manufacturing

- Ordering PCBs
- PCB Manufacturing Process

Home Automation Board PCB Design

- Introduction
- Home Automation Board Schematic
- Home Automation Board PCB

Soldering

- Basics of Soldering, Soldering equipment
- Through-Hole soldering
- SMT Soldering
- Cleaning your soldering iro

course projects

- PCB Designing and Soldering LED Circuit
- PCB Designing of Home Automation Board

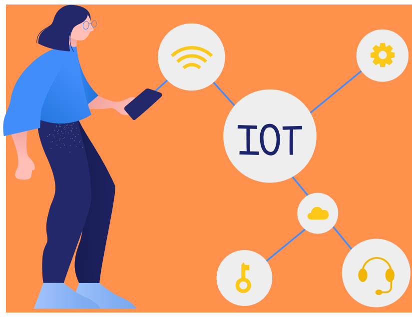
Roadmap for Internet of Things(IoT)

Engineering 2nd Year

Python Programming
IoT using Raspberry Pi / ESP32 and Python Course

Minor Project
Hackathon
Internship

Engineering 3rd Year



IoT using Arduino
IoT Cloud Platforms
IoT Data Analytics (OR)
PCB Designing and Soldering

Hackathons
Internship
Major Project
Core Placements

Certifications

HackerRank

ARDUINO

LinkedIn

IABAC International Association of Business Analytics Certification

CISCO

NVIDIA

aws

Azure

Fee

INR 13990/-

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